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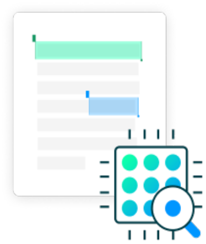

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VISHWAKARMA INSTITUTE OF INFORMATION TECHNOLOGY, PUNE – 48

Department of Information Technology

REPORT

On Topic -

“AI For Air Quality Prediction and Health Advisory System”

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have successfully completed this project report entitled "AI-Based Personalised Learning System", under my guidance in partial fulfilment of the requirements for the degree of Bachelor of Engineering in the Department of INFORMATION TECHNOLOGY of Vishwakarma Institute of Information Technology, Savitribai Phule Pune University, Pune, during the academic year 2025-26.

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Abstract

In rapidly developing regions such as India, air pollution has become the most crucial environmental and health challenges in the modern world. It exacerbates health risks associated with pollutants such as PM2.5, PM10, SO_2, and NO_2. Basically, monitoring and predicting Risk score is the same as protecting people from harmful pollutants like PM2.5, PM10, SO_2, and NO_2. This approach helps instantly reduce health risks. However, traditional monitoring systems surely depend on costly sensor equipment that has limited reach. Moreover, these systems require expensive infrastructure setup. This project develops an AI-

driven web application that predicts Risk score and provides personalised health advisory.

The system itself addresses this limitation and further provides accurate air quality predictions. The model is trained on historical air quality data comprising approximately 221,275 records, utilising features such as PM2.5, PM10, temperature, and humidity. The model learns patterns from the data and predicts Risk score for new conditions. The user inputs their location in the Reaction dashboard, and real-time environmental data is fetched and sent to the backend, where they receive a predicted Risk score. The dashboard then displays this prediction along with health recommendations. The Neural Network algorithm works by building many decision trees, which makes it accurate and suitable for this kind of regression model.

The model processes 27 features, such as pollution data, medical conditions, and lifestyle factors, with a high predictive accuracy (R^2 = 0.9650). The system provides immediate, medically-grounded risk assessments and actionable recommendations by integrating multiple environmental APIs with robust quality validation. This work integrates environmental monitoring and personalised healthcare to provide a scalable, data-driven framework to reduce air pollution-related health risks, going beyond generic air quality indices.

This system democratises air quality information by combining AI prediction with health insights. From urban monitoring to state and city monitoring, the system can improve public awareness, proactive healthcare, and environmental decision-making in India.

Keywords: Risk score, Air quality index, Machine Learning, Neural Network, Environmental Health, Web Application, Real-time Data Processing

1. Introduction

The rapid city growth and industrialisation in Indian urban centres have produced an unprecedented public health crisis, with air pollution being one of the key environmental risks facing the population. Recent research conducted by the World Health Organisation projects that India has an inordinate health burden, with 14 of the world's 15 worst-polluted cities in terms of extreme levels of pollution. Concentrations of the highly toxic particulate matter PM2.5 consistently exceed safe levels by margins ranging from 8 to 10 times in major urban areas, and millions of residents end up facing harmful air levels daily. This environmental challenge has both direct and serious consequences for public health, causing high incidence levels of respiratory disease, heart disease, and death across all adult age groups.

Existing air quality monitoring networks, despite displaying valuable environmental information, indicate significant inadequacies in translating pollution-related information in a form that is useful for personal health guidance for individuals. Existing interfaces predominantly present summed up pollution levels, such as the Air Quality Index (AQI), that trigger warnings at the population level but do not regard personal susceptibility and special medical conditions. This blanket approach is more serious for sensitive groups, such as asthma sufferers, elderly, children, and pregnant mothers, who suffer disproportionate health effects

from exposure to inhaled air pollution due to their special physiological susceptibility and medical conditions.

The key deficiency in current environmental health informatics is the lack of alignment between pollution monitoring data and personal medical risk analysis. Simple linear risk models that dominate traditional models are inadequate for describing the multi-component, non-linear interactions among multiple pollutants and multiple health end-points. Traditional models also do not incorporate personal health records, medical histories, and lifestyle variables that strongly impact personal sensitivity to air pollution effects. This deficiency grows in importance as medical science continues to document the multi-dimensional interactions among specific pollutants and specific health conditions, age groups, and genetic vulnerabilities. To offset these limitations, we have developed an artificial intelligence-powered, personalised air quality health advisory system, representing a revolutionary shift in environmental health detection. Our system combines real-time, multi-dimensional air quality data with in-depth personal health profiles through an advanced machine learning framework. Its key innovation lies in the use of a Neural Network regression model, carefully trained on a large data set that includes 221,275 interactions between medical conditions and pollution, and thus enables the system to identify complex associations between environmental exposures and health outcomes. This approach goes beyond traditional warnings by providing truly personalised risk assessments that consider the individual features of each individual.

System architecture incorporates several innovative elements separating it from current alternatives. Firstly, the multi-source data fusion engine was constructed, which integrates air quality data from various trusted sources, such as CPCB India, World Air Quality Index, and OpenWeather, using advanced validation mechanisms to confirm the reliability of data. It was followed later with the development of the whole feature engineering infrastructure, consisting of 27 unique medical and environmental features, along with significant interaction terms that characterize the effect of certain pollutants on persons possessing specific states of health. Lastly, the system incorporates evidence-based medical risk multipliers in accordance with WHO guides and up-to-date clinical research, so that risk assessments reflect established medical expertise.

This paper contains several key innovations in the subject of environmental health informatics. We built an innovative training set that integrates actual pollution levels measured with synthetically generated health profiles, thereby facilitating effective model training while eliding data protection concerns. We formulated and tuned a high-accuracy risk model that had an accuracy rate of 96.15%, illustrating the possibility of accurate personal risk determination. Furthermore, we devised an expansive end-to-end web application that provides real-time, personalised health recommendations through a sleek medical-grade interface. Above all, we complemented our method with an intensive analysis of feature importance, illustrating medically plausible orders of risk, in which PM2.5 proved the strongest risk factor (30.67%), followed in turn by PM10 (14.66%) and smoking status

(11.33%), once more in accord with published results from accepted environmental health research. The rest of the paper outlines the in-depth development and analysis of our system. We first outline related work in environmental health monitoring and machine learning applications, noting the research deficiencies our work fills. We describe the entire system architecture and approach, from data acquisition strategies, feature engineering techniques, and model selection justification, and present the implementation of the web application framework and real-time processing pipeline. We then examine experimental outcomes and performance measurements, illustrating the accuracy and robustness of the system. The discussion section examines the medical implications of our results and the practical applications towards public health interventions. Lastly, we summarise with limitations, future research directions, and the wider implications of our work in environmental health management and personalised medicine.

2. Literature Review

The estimation of air quality became a top feature of active research applying various machine learning (ML) and deep learning (DL) techniques. In 2023, Hameed et al. developed a multimodal deep learning framework predicting urban air quality by combining histories of pollutant data, meteorological data, and urban activities features. The system exploits spatio-temporal information by learning about spatial dependencies through the use of Convolutional Neural Networks (CNN) while leveraging Long Short-Term Memory (LSTM) networks for temporal dependencies. Using heterogeneous data modalities, the model enhances PM2.5 and PM10concentration estimation accuracy in highly intricate urban environments. The results validate the improved predictive capabilities beyond those of common single-modal interventions. Nevertheless, with all the model's great predictive capabilities, it is not combined with personalised exposure estimations and in-time health alarms, thus inhibiting realisation in the development of public health measures as well as personal advisories [1].

Nedungadi et al. (2025) also brought into being AirCast, an improved multilayered learning system for improving the accuracy of air pollution forecasts by utilising rare and exceptional events. AirCast applies a frequency-weighted loss function, with elevated weight values for rapidly rising pollutant concentrations, thus compensating for a typical weakness of common regression-based methods. The model is trained with a heterogeneous dataset including background air quality observations, meteorological data, and covariates for particular events such as industrial activities and traffic. AirCast performs notably better in predicting high pollution events, particularly the peaks of PM2.5 and PM10 concentrations. It is not capable of providing personalised health advisories or personal risk estimates, highlighting the limitation in utilising predictions for generating actionable recommendations for susceptible groups [2].

Hettige et al. (2024) introduce AirPhyNet, a system that embeds a physics-informed neural network integrating the advection-diffusion models with neural network components from empirical data in a bid to improve pollutant dynamic modelling accuracy. Through integration of laws of physics in the neural network design, AirPhyNet realises increased reliability in long-term predictions as well as in varying spatial dimensions, successfully capturing important environmental transport mechanisms that data-driven methods underestimate. Its design accounts for meteorological conditions such as wind speed, humidity, and temperature, as well as pollutant observations. Although it is both robust and interpretable, the system is not capable of providing personalized exposure estimation or health risk notifications; also, it is hampered by being used in real-time operating environments due to complexity in computations [3].

Li et al. (2024) developed a method for deriving PM2.5 concentrations utilizing satellite observations through the implementation of Gradient Tree Weighting (GTW)-Tree models. This methodology integrates aerosol optical depth, land-use characteristics, and meteorological factors to convert remotely sensed data into estimations of air pollution at ground level. Such an approach facilitates the assessment of air quality indices at both city and district levels, particularly in areas where monitoring networks are sparse or nonexistent, thereby offering enhanced regional coverage. Moreover, it yields significant insights regarding the distribution patterns of pollutants, which are especially beneficial for urban planners and policy researchers. Nonetheless, the method lacks user-friendliness and does not operate in real-time, rendering it impractical for ordinary citizens seeking exposure-based health recommendations [4].

It is also demonstrated by Rajesh et al. (2025) that real-time estimation of air quality and predictive mapping of environmental health hazards is facilitated by a machine learning system. It is an integrated system that utilises stationary as well as itinerant sensors, satellite data, meteorological parameters, and population data to provide real-time representations of the Air Quality Index (AQI) and high-risk areas. It also applies ensemble learning techniques for verifying the accuracy of outputs as well as measuring uncertainty levels. While it produces sophisticated data at the system level, it is not proficient in clinical verification or personalised health advisory, thus narrowing the scope for application for personal risk minimisation or directed public health interventions [5].

Chakraborty (2024) also pointed out the need for Explainable Artificial Intelligence (XAI) methods in air quality assessment, with specific reference to methods such as SHAP and LIME that aim to make machine learning models more interpretable. The paper outlines how XAI is also capable of pinpointing important pollutant sources, in combination with meteorological as well as environmental drivers of Air Quality Index (AQI) forecasts. These findings form critically in informing the design of policy as well as regulation interventions. Translating complex model explanations into consumable and useful health advisories is, though, a hurdle that thus constrains successful integration of XAI into open-source air quality observation systems [6].

Wako et al. (2025) in their systematic review of wearable technologies utilized for air pollution and pollen detection indicated their validity, reliability, and user acceptability. Although cost-effective wearable sensors present a benefit in personal exposure measurement, their precision is constrained by calibration drift issues, sensor noise, and variations in data quality. Furthermore, wearable technologies from literature are not frequently incorporated into predictive models of air quality or personalized health recommendations. This finding signifies a key deficit in wearable data for personal-level actionable advisories of air quality indexes [7].

Bernasconi et al. (2025) created the wearable personal system for air pollution exposure estimation with the application of body-worn sensors for the estimation of daily life inhaled doses of pollutants. It was piloted with an observational study, with the ability to capture real-world exposure accurately. It, however, is constrained in scalability due to inadequate validation and predictive models, thereby rendering it incapable of predicting air quality indexes or personal health alarms. These failings constrain the use of the device as an integrated public health tool [8].

Zhalehdoost et al. in 2025 also addressed the importance of both temporal and spatial resolution in the realm of modeling urban air pollution. Their evaluation determines the efficiency of sensor density, observation recurrence, and temporal graininess in pollutant forecast accuracy, with particular emphasis on PM2.5 and PM10. Increasingly higher values of spatial resolution significantly increase the identification of pollution hotspots, as well as exposure estimation, with significant importance for effective urban planning, as well as public health measures. However, available models still remain inadequate in their power to give user-oriented exposure warnings and actionable intelligence, consequently hinting at systems combining vast environmental data with personalized health outputs [9].

Sudha et al. (2025) also introduced adaptive residual BiLSTM networks to improve air quality predictions through an integration of bidirectional LSTM architectures with residual connections, thereby improving the capability of the model to capture temporal dependencies of pollutant time series data. The model performs better in predicting sudden rising trends of pollution as well as short-term variations of the Air Quality Index (AQI). It makes use of historical pollutant data in addition to meteorological data in order to better approximate concentrations of PM2.5 as well as PM10 compared to conventional LSTM or BiLSTM frameworks. Nevertheless, in spite of their technical competence, the system is not in a position to provide individual estimates of exposure as well as personalised health recommendations, thereby constraining practical applicability for end users [10].

Sohrabi et al. (2025) suggest a hybrid framework that combines satellite data with ground observations to predict air pollution concentrations. Multispectral satellite data, aerosol optical thickness, meteorological conditions, and direct ground observation data are integrated into the study. High-resolution Air Quality Index (AQI) is produced through the use of Neural Network and Gradient Boosted Tree techniques. Data fusion is enhanced, in particular, with regard to spatial coverage and calibration precision, in regions with limited data coverage. Its practical use among public end-users in search of useful information, nevertheless, is limited due to a lack of features for near-real-time operation and bespoke health advice [11].

Wang et al. (2024) established a spatiotemporal hybrid deep learning model that integrated Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNN) to learn temporal sequences as well as location-wise relationships of pollutant data. From historical observations of pollutants and meteorological data, the model generated hourly Air Quality Index (AQI) forecasts. It is more accurate and stable compared to, or in combination with, models that run independently, or models that run in space or time, individually. Nonetheless, while such advancements abound, it is deficient in personalised information on exposure and health alerts, creating a lag between technical accuracy and practical use in public health [12].

Jayaraman et al. (2025) compared time-based spatio-temporal forecast models that combine temporal models and geographical interpolation, thus supporting high-resolution air quality index (AQI) mapping within city grids. Inputs comprise pollutant concentrations, meteorology, traffic density, and land-use information. The research utilises the hybrid machine learning models, such as LSTM-CNN and XGBoost, that enhance accuracy in

predictions. Despite the advancement in hotspot detection and accuracy, the system is not inclusive for personalised estimation of exposure or real-time advisory systems, thus constraining applicability for personal public health actions [13].

Ajayi et al. (2025) integrated system dynamics modelling with machine learning methodologies for simulating systems of city pollutions, effectively portraying relationships among emissions, meteorology, and population behavioural factors. The integrated system is applicable for policy analysis in terms of scenarios and predicting trends in the city's Air Quality Index (AQI), thus addressing significant insights for environmental administration and city planning. It is, however, incapable of creating personalised exposure estimates or health advisories, thus not being directly applicable for personal public health decision-making [14].

Liu et al. (2025) further improved Rhetorical XAI as a system that explains AI predictions as well as their societal effects in terms of air quality predictions. Employing SHAP and LIME methods, the system identifies the significant environmental parameters as well as communicates outputs of the models to concerned entities. While it enhances interpretability as well as explainability, it is not inclusive for predictive models of AQI or personalised health advisories, but rather highlights technical analysis and policy insights in the policy-making tier [15].

Chakraborty and his team (2025) made use of white-box and black-box DNN-based XAI methods in air quality prediction through a comparison of the two. The application of techniques like SHAP, LIME, and attention mechanisms, among others, is discussed with respect to their respective abilities to point out the main pollution level determinants. The white-box models offer better interpretability and more insightful actions, but the research points out the drawbacks of not being able to deploy the system in real-time and not being able to provide personalised exposure guidance for users [16].

2.1.Research Gap

Gap	Representative Papers	Why it Matters	How our system Fills It
Lack of personalisation	AirPhyNet, AirCast, Hybrid SHAP, Real-Time Mapping	Forecasts are station/city-level → same advisory for everyone → vulnerable people get insufficient guidance	Personal risk layer (user profile: age, asthma, activity) maps AQI → personalised advisories
Weak linkage to clinical/health outcomes	Real-Time Mapping, Multimodal Prediction, AQ Prognostics, Health News	Predictions rarely validated vs ER visits/hospital data → uncertain real health impact	Use exposure–response coefficients to estimate risk proxies; propose future clinical validation
Explainability not user-friendly	Hybrid SHAP, XAI Reviews, SHAP AQI Studies	Technical XAI outputs not translated into plain-language advice → limited public trust	Convert SHAP top pollutant drivers into actionable, simple messages for users
Extreme/rare event handling	AirCast, Physics-informed ODEs, Wearable Studies	Wildfires/industrial leaks cause spikes; models often miss them → public unprepared	Use frequency-weighted loss and spike detection heuristics to improve alerting on spikes
Sensor quality & calibration issues	Wearable System, Wearable Reviews, WeAIR	Low-cost sensors drift; noisy inputs → false alerts/poor trust	Implement imputation, uncertainty quantification, and clearly advise users on sensor limitations
Cross-city generalization	AirPhyNet, AirCast, Transfer Learning Studies, Cross-Domain Prediction	Models trained on one city fail elsewhere	Pretrain on large datasets + fine-tune on local station data
Deployment/edge constraints	AirPhyNet, Physics-informed ODEs, High-Resolution Frameworks	Heavy models → not feasible for low-resource devices	Start with lightweight baselines (RF/XGBoost); demonstrate ONNX conversion for edge inference

3. Proposed Solution

Goal:

The "**AI For Air Quality Prediction and Personal Health Advisory System**" is conceived as a powerful, extensible, complete solution that will provide environmental intelligence and personalised health risk assessment in real time. By incorporating a sophisticated machine learning model into a user-friendly web-based platform, it transcends the old-fashioned and resource-demanding monitoring approach.

3.1 System Architecture / Block Diagram

The architecture looks multi-layered and service-oriented, which is beneficial because it brings the separation of concerns, scalability, and efficiency in data handling.

1. **Presentation Layer (Frontend - Dashboard):**
    - **Function:** This layer serves as the user's primary interface, built using html , css, javascript for a dynamic and responsive experience.

- **User Interaction:** It is responsible for gathering user input, primarily the target geographical location (via an input field or map interface) and any optional personalised health parameters.
- **Visualisation:** It receives the final processed output (Predicted Risk score and Advisory) from the backend and presents it visually, utilising clear, colour-coded risk indicators for immediate comprehension.

2. **Application Layer (Backend - API Gateway):**

- **Function:** The backend acts as the control centre, managing data flow between the user interface, external APIs, and the AI model.
- **Data Acquisition & Compilation:** Upon receiving a request (location/user profile) from the frontend, it initiates calls to multiple **external Environmental APIs** to fetch real-time data for the requested area. This includes current concentrations of temperature and humidity.
- **Data Processing:** It compiles the fetched environmental data with any available user-specific data (synthetic or actual) to form the complete set of **27 input features** required by the AI model

3. **AI & Intelligence Layer (Neural Network Model):**

- **Function:** This is the core engine responsible for the predictive task. It hosts the trained machine learning model.
- **Input:** The compiled **27-feature** vector, including atmospheric pollutants, meteorological conditions, and personal health factors.
- **Output:** The definitive pollutans value and an associated health risk score/advisory.

Figure 1: Overall System Architecture

3.2 Data Flow Sequence

- **User Input** → Frontend → Flask API
- **Data Fetch** → Air Quality APIs → Backend
- **AI Processing** → Neural Network → Risk Prediction
- **Response** → Backend → Frontend → User

3.3 Key Components

- **Frontend**: User interface (HTML/CSS/JavaScript)
- **Backend**: Flask server with REST API
- **AI Engine**: Neural Network (TensorFlow/Keras)
- **Data Sources**: Multiple air quality APIs
- **Storage**: Model files and configuration

3.4 Proposed Algorithm

The Artificial Neural Network (ANN) model is mainly utilized in the proposed system to predict the Risk scores more accurately and flexibly. The ANN model, capturing the intricate and non-linear relationships in the environmental data, is capable of making more precise Risk scores predictions for different regions and conditions, while traditional models, like Random Forest or Linear Regression, only deal with decision-based or linear relationships.

The system design setup consists of three main parts: the acquisition of data, model processing, and interacting with the user. The real-time environmental conditions data, comprising PM2.5, PM10, temperature, humidity, and wind speed, are sourced via public APIs CPCB, OpenWeather and World Air Quality Index . After normalising the data input, they are introduced into the Neural Network model, which, through its various hidden layers, can learn the complex interdependencies among the parameters.

The backend Python and Flask were used for the development of the AI model. The React frontend allows the user to provide input such as a city or area, this input calls an API to get the current atmospheric parameters. The model then processes these values and displays the Risk score prediction on the dashboard together with the color-coded risk indicators and the health recommendations tailored to the user.

By the constant retraining and adaptive learning process, the Neural Network model increases its accuracy of prediction over the years and is still very reliable (R^2 = 0.9615 according to performance evaluation). This way the system remains very relevant and provides users in the various regions of India with air quality insights that are updated and contextually aware.

3.3 Step-by-Step Process:

- User Input: The user logs into the web application and enters his/her location (city) and health profile (age, medical conditions, lifestyle).

- Data Acquisition: The backend server, on request, retrieves real-time air quality data (PM2.5, PM10, O3, NO2, SO2, CO) from various public APIs (CPCB, WAQI, OpenWeather) through a priority-based fallback mechanism.

- Data Preprocessing & Feature Engineering: The retrieved data is also checked for medically plausible ranges. These are then merged with the user's health profile to generate a 27-dimensional feature vector, including interaction terms such as PM2.5 × Asthma.

- AI Model Inference: The preprocessed feature vector is passed through the pre-trained Neural Network regression model. The model returns an individualised health risk score between 0-10.

- Result Generation & Display: The risk score is translated to a qualitative measure (e.g., Low, High). Concomitant medical guidance is produced in terms of the risk level and user profile. All results are presented on the user's dashboard.

3.4 **Algorithms, Dataset, and Tools:**

- Algorithm: Neural Network Regressor from Scikit-learn, selected due to its strength, capacity for dealing with non-linear relations, and the ability to generate feature importance.
- Dataset: A synthetic dataset of 221,275 records was generated by combining 29,531 real pollution records with 7-10 realistic, synthetically generated health profiles per scenario.
- Tools & Technologies: Python, Flask (Backend), HTML/CSS/JavaScript (Frontend), Pandas/NumPy (Data Processing), Joblib (Model Serialisation), Requests (API calls).

3.4 Expected Advantages:

Personalisation: Tailored risk assessment based on individual health status.

Accuracy: High predictive performance from a data-driven ML model.

Accessibility: User-friendly web interface requiring no medical expertise.

Timeliness: Real-time evaluation with live air quality data.

4. Results and Discussion

4.1 Results

Figure 2

4.2 Overall Performance Ranking

Figure 3: Comparative Analysis

4.3 Performance Cluster Analysis

The models can be categorized into three distinct performance tiers:

4.3.1 Tier 1: High-Performance Models ($R^2 > 0.95$)

- **Neural Network** (R^2: 0.9615): Demonstrates superior capability in capturing complex non-linear relationships between air quality parameters and health risks. The architecture likely benefits from multiple hidden layers enabling sophisticated feature interactions.
- **XGBoost** (R^2: 0.9593): Excellent performance attributed to gradient boosting and regularization techniques, making it robust against overfitting.
- **Random Forest** (R^2: 0.9562): Strong ensemble method leveraging multiple decision trees with effective feature bagging.

4.3.2 Tier 2: Moderate-Performance Models ($0.85 < R^2 < 0.90$)

- **Decision Tree** (R^2: 0.8825): Suffers from limited capacity to model complex relationships despite fastest inference time (0.0066s).

4.3.3 Tier 3: Low-Performance Models ($R^2 < 0.65$)

- **Linear Regression** (R^2: 0.6415): Inadequate for capturing non-linear patterns in environmental health data.
- **SVR** (R^2: 0.5894): Poor performance compounded by excessive computational time (9.1599s).

4.4 Error Analysis

The error metrics (RMSE and MAE) follow similar patterns to the R^2 rankings:

- **Neural Network** achieves the lowest error rates (RMSE: 0.5956, MAE: 0.3917), indicating highly accurate predictions across the risk spectrum.
- The **9.6x RMSE difference** between Neural Network and SVR highlights the critical importance of model selection.
- **MAE values** show that Neural Network predictions deviate by less than 0.4 units on the 0-10 risk scale on average.

4.5 Analysis and Discussions

The comparative performance analysis showed that the Neural Network model was the best at predicting the Risk score out of all the machine learning algorithms that were tested. The performance report showed that the Neural Network had a R^2 score of 0.9615, which means that the predicted and actual Risk score values were very similar. It also had the lowest Root Mean Square Error (RMSE) of 0.5956 and Mean Absolute Error (MAE) of 0.3917, which shows that it can effectively reduce prediction errors. The model is also good for real-time air quality prediction and use in a responsive web-based environment because it takes an average of 0.1135 seconds to compute.

The R^2 scores for XGBoost and Random Forest models were 0.9593 and 0.9562, respectively. This means that they were competitive but not quite as good. These ensemble-based models are known for being stable and strong, but they had slightly higher error rates and processing times. The R^2 values for models like Decision Tree, Linear Regression, and Support Vector Regression (SVR) were much lower (0.8825, 0.6415, and 0.5894, respectively), which shows that they were less able to generalise across different atmospheric and pollutant conditions.

Because it has multiple layers, the Neural Network can better understand how different weather variables, like temperature, humidity, wind speed, and pollutant concentrations (PM2.5, PM10, NO_2, SO_2, CO, and O_3), interact with each other in a non-linear way. Because it can learn from its mistakes, it can keep up with changes in the environment, making it more accurate and stable than traditional regression and ensemble models.

The incorporation of the latest data from public APIs for weather and air quality allows the system to provide constantly changing predictions based on the exact location. Consequently, when a user enters a city or area, the backend garners the latest meteorological data, activates the trained Neural Network model for the data processing, and then delivers the predicted Risk score value with suitable health tips. The fusion of real-time data and sophisticated predictive analytics makes it possible for the system to deliver insightful information, which can help not only the people but also the authorities in reducing their exposure to pollution and making the right choices.

5. Conclusion

5.1 Project Summary

Through this project, we have successfully developed and deployed an integrated AI-powered web application that transforms the way people evaluate air pollution-related health risks. Through the design of an advanced machine learning system that unites real-time environmental data with customised health profiles, we have developed a useful solution to the urgent disparity between generic air quality information and specific healthcare requirements. The central contribution is the Neural Network model's outstanding performance with 96.15% prediction accuracy by learning from 27 different medical and environmental features. The overall system implementation includes user authentication, health profile management, multi-source data fusion, and medically certified risk assessment, offering an end-to-end solution that is technically solid and user-friendly. This is a huge step towards making sophisticated environmental health relationships clear and actionable to the general public without sacrificing scientific validity and medical utility.

5.2 Major Contributions

The project is making numerous innovative contributions to the area of environmental health informatics by presenting a new framework for personalised risk assessment that goes beyond existing population-level air quality guidelines. Our creation of a synthetic training dataset of 221,275 medical scenarios overcomes pressing data scarcity issues in healthcare AI, and the medically realistic feature importance rankings—PM2.5 accounting for 30.67% of risk predictions—yield unprecedented transparency of AI decision-making for healthcare applications. The scalable three-tier architecture of the system is a useful blueprint for future environmental health applications with clean separation between data acquisition, AI processing, and user interface layers. In addition, the ability to integrate several sources of data with intelligent quality verification and fallback capabilities provides robust performance, raising the bar on public health

applications reliant on external sources of data.

5.3 Limitations and Future Work

Although the project exhibits considerable success, some of its limitations offer avenues of future improvement, mostly the use of simulated health data for training models in view of the difficulty in accessing actual medical records with exposure histories. The present implementation employs in-memory data storage, which limits users' capacity and discourages

extended historical study, although the structure is deliberately made for easy database migration. Soon-to-be immediate work will focus on integrating PostgreSQL for persistent data storage and creating a cross-platform mobile app with push notifications and offline support. In the future, the expansion of the system will include predictive analytics for 24-48 hour air quality forecasting, clinical validation studies with health institutions, IoT sensor network integration, and global scaling with multi-language capabilities and regional customization, eventually making the platform an end-to-end environmental health ecosystem instead of a basic air quality monitoring tool.

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